

VAT Identifier Discovery Research and PoC

Part 1: Research

I started by looking at the easiest ways I could find VAT numbers for UK companies without paying for a database.

The first thing I checked was the HMRC VAT checker mentioned in the challenge resources. It is very useful for checking a VAT number once you already have one, but it does not let you search for a company's VAT number using its name or company number. Because of this, I could use it to confirm numbers, but not to find them in the first place.

I also used Companies House as the main source for the companies themselves. It gives me the company name, company number, address, status, SIC code and other company information, but it does not give the VAT number.

I then tried looking for VAT numbers manually through Google. This actually worked quite well for individual companies. In some cases I could find a VAT number on a company website or another website and then check it with HMRC.

The problem was the amount of work involved. For one company this is fine, but doing the same thing for hundreds or thousands of companies would take a very long time. I also had to decide whether a result really belonged to the company I was looking for.

I then tried finding the official website of each company first and searching there. This did not work as well as I expected because many companies did not have an obvious website. Some of the first results were also directories or other websites instead of the actual company website.

After that I tried automatically opening websites with Python using requests and BeautifulSoup. This ran into a different problem. Several websites returned HTTP 403 or 406 errors when I tried to access them, and some other pages did not contain any useful VAT information. This made the basic approach of downloading pages and searching them automatically unreliable.

I also tried the Brave Search API. I used several different searches for the same company, including searches for the company name together with "VAT", "VAT number" and "VAT registration". The results were inconsistent. In one case I already knew that a VAT number existed because I had found it manually, but the API still did not find it. This made me unsure that I could trust this method for a larger dataset.

I looked briefly into finding public invoices and other documents as another possible source. I did not get enough useful results from this to make it worth continuing with.

I also tested a Gemini API approach that could search Google and return a possible VAT number and its source. I could not properly test this because the API repeatedly returned quota errors. Because of this, I did not use it for the final results.

The two commercial services that looked most useful were Endole and VAT-Search. Both already have company information together with VAT information, which solves most of the searching problem. The downside is that the free versions are very limited and useful larger-scale access requires a subscription. The prices I found were around £38/month for Endole and around €9 for the cheapest VAT-Search package, which only allows a small number of searches.

One of the things I found most surprising was how difficult it is to get this information for free. You would expect company information that is publicly relevant to be easy to search, but the VAT number is often not available through a simple company-name lookup.

I was also surprised by how much companies charge for something that sounds simple on the surface: finding the VAT number for a company. After testing the free options, I can understand better why these services have value.

Overall, I found that there is no simple free source that I could use to find VAT numbers for a large number of companies. Manual searching works reasonably well for small numbers, but it becomes too slow. Automatic crawling had problems with websites blocking access, and search APIs did not give consistent enough results in my tests. Commercial services looked much more practical, but they require payment.

Update

I then tried using Selenium to automate company searches directly on Endole's website. This did not work because the website repeatedly asked for human verification, and I was not able to get the search working reliably through Selenium.

I then tried Google searches through a Selenium-controlled browser instead. This worked much better. The browser was able to load the normal Google results, and in some cases Google's AI-generated result also provided a VAT number or information that the company did not have a publicly visible VAT number.

For the first test, I used 100 active companies with SIC code 45112, incorporated between 2014 and 2019.

```
SELECT *  
FROM vat_research  
WHERE incorporation_date BETWEEN '2014-01-01' AND '2020-01-01'  
AND status = 'Active'  
AND sic_code LIKE '45112%'  
ORDER BY random()  
LIMIT 100;
```

The automated Google search found a possible VAT number for 30 out of the 100 companies. The process was much slower than a normal bulk API search, but it was still considerably faster than searching manually.

The search did occasionally trigger Google's unusual traffic verification. This happened only once during the 100-company test, after around 88 searches. After completing the verification manually, the search could continue.

Some searches also took longer because Google's AI result needed more time to finish. In a few of these cases I manually checked the companies afterwards and confirmed that there was still no publicly available VAT number.

For the second test, I selected 500 active companies incorporated from 2020 onwards and excluded SIC code 45112 so that the larger test would use a different group of companies.

```
SELECT *  
FROM vat_research  
WHERE incorporation_date > '2020-01-01'  
AND status = 'Active'  
AND sic_code NOT LIKE '45112%'  
ORDER BY random()  
LIMIT 500;
```

The next step was to check the VAT candidates against HMRC automatically using Selenium.

This produced another important finding. A VAT number can be valid according to HMRC but still belong to a different company. For example, one candidate for CROFT CAR SALES LIMITED was confirmed by HMRC as a valid VAT number, but HMRC showed that the number actually belonged to WHYTE CROFT CAR SALES LTD at a different address.

This means that simply checking whether a VAT number is valid is not enough. The company name and address also need to be compared with the company being researched. A valid VAT number belonging to the wrong company is still a bad result, since it could lead to the wrong company being matched in the final dataset.

During the HMRC checks I also encountered rate limiting. This means that the public HMRC checker cannot simply be queried indefinitely without pauses. I adjusted the script to wait before trying again when this happened.

Part 2: Proof of Concept

For the proof of concept, I wanted to test the process on real companies instead of starting with companies where I already knew the VAT number.

I imported a large Companies House dataset into PostgreSQL and selected 100 active UK companies. I limited the sample to companies with SIC code 45112, which is the code for the sale of used cars and light motor vehicles, and companies incorporated between 2014 and 2019.

This is not a representative sample of all UK companies. I chose it mainly because it gave me a clear group of real companies that I could test in the same way.

I manually checked the first 10 companies from the sample.

For these 10 companies, I found 3 possible VAT numbers. I then checked all 3 with HMRC. 2 were confirmed and 1 was rejected.

So, in this small test:

- 10 companies were checked
- 3 possible VAT numbers were found
- 2 were confirmed by HMRC
- 1 was a false positive

This means that I found a possible number for 30% of the companies I checked. One third of the numbers I found were wrong.

I would not treat these numbers as the final accuracy or coverage of the method because the sample is very small and was not randomly selected from all UK companies. They are mainly useful for showing what happened in the first test.

One useful example was Vogue Competitions. I found a VAT number through Endole and then confirmed it with HMRC.

I also found a VAT number that looked valid in search results but was rejected by HMRC. This was important because it showed why I cannot simply take any VAT number that appears online and add it to the database. Indeed, a wrong number could be worse than having no number at all.

The next step is to test the same general idea using Selenium instead of simple requests and BeautifulSoup. The main reason for this is that my first crawling attempt was blocked by several websites with 403 and 406 responses. Selenium lets me test whether opening the pages in a real browser gives better results.

For the larger test, I will keep the same basic measurements: how many companies I managed to check, how many possible VAT numbers I found, how many were confirmed by HMRC and how many were rejected.

An important limitation is that not finding a VAT number does not prove that the company is not VAT registered. It only means that my search did not find one.

Update

For the actual proof of concept, I used the Google + Selenium search method described above on 100 companies from the first sample.

The search found a possible VAT number for 30 of the 100 companies.

I then checked the candidates against HMRC. I did not treat a VAT number as correct just because HMRC said it was valid. I also compared the company name and address returned by HMRC with the company from my sample.

This revealed an important type of false positive. One VAT number was valid according to HMRC, but it belonged to a different company with a different address. I marked these cases as needing further checking rather than accepting them automatically.

During the HMRC checks, the public checker also started returning HTTP 429 responses after a number of requests. I added delays and retries, but this still showed that the public checker itself could become a limitation when the number of companies increases.

Update 2

The 500-company test has now been completed.

I used the same Google + Selenium search method on 500 randomly selected active companies incorporated from 2020 onwards, excluding SIC code 45112.

The search returned at least one possible VAT number for **43 out of 500 companies (8.6%)**.

This is noticeably lower than the 100-company test, where 30 candidates were found, but the two samples are different and I would not use the results to estimate general UK VAT coverage.

The next step is to check all 43 VAT candidates, alongside the 31 from the first test, against HMRC. This is important because the Google search only gives possible candidates, and some of them may be invalid or may belong to a different company.

The HMRC checker has also started returning a significant number of HTTP 429 errors during automated checking. I have added delays and retries, but the rate limiting is still a limitation of using the public checker at this scale.

I will continue the HMRC checks once the rate limit allows it and record how many candidates are actually verified, how many are false positives, and how many still need checking. In the worst case, if HMRC rate limiting prevents reliable automation, all candidates could still be checked manually. There are currently 76 candidates across the two tests, so this is still manageable for the proof of concept, although it would become impractical at much larger scale.

Update 3

While checking the results more closely, I found another problem with the Google search method.

At first I was using the company name and company number in quotes in the search query. I expected this to make the results more accurate, but in some cases it actually made the search worse. Google can still return VAT numbers that are related to a different company.

One example was especially interesting because the same VAT number appeared as a candidate for five different companies in the sample. I manually checked one of these cases and found that the VAT number actually belonged to a completely different company.

This is an important limitation of the current approach. Finding a VAT number in Google results does not mean that the number belongs to the company being searched. The candidate still needs to be checked against HMRC, including the registered company name and address.

The HMRC checks worked for the first group of candidates, but after that the public checker started returning HTTP 429 responses consistently. This means that the checker was rate limiting the automated requests. I added delays and retries, but at this point I will need to manually check some of the remaining candidates if the rate limit does not clear.

Out of the 76 candidates found across both samples, the final verification results are:

- Candidates checked through HMRC: **76**
- Confirmed as belonging to the correct company: **24**
- FALSE POSITIVE: **3**
- NEEDS CHECKING: **25**
- Total HTTP 429 rate limit rejects: **27**

The main limitation of my current approach is that I am relying on a single discovery method. Google search can find useful information, but it is not reliable enough on its own to establish a company-to-VAT relationship. The results can be incomplete, overly broad, or associated with another company.

This also explains why commercial services such as Endole are more useful for this type of problem. They can combine information from multiple sources instead of relying on a single search method.

The other major limitation is the HMRC verification step. The public checker works well for confirming a candidate, but automated use is affected by rate limiting. HMRC also provides an API, but access requires registration and verification, which can take around two weeks. I did not have enough time to obtain access during this proof of concept.

While trying to manually verify the remaining results marked as **NEEDS CHECKING**, **RATE LIMIT**, or even **FALSE POSITIVE**, I encountered another roadblock.

HMRC had fully rate limited my access to the VAT checker by this point and was returning **HTTP 429** for every search. Because of this, I was unable to manually check the remaining VAT numbers and see the full results.

This means that I could not confirm whether some of the remaining candidates were actually valid, belonged to another company, or were simply false positives. I therefore had to leave these results unverified instead of making assumptions about them.

Part 3: What I Would Do With Real Resources

With a normal laptop and no budget, I have already run into quite a few limits. With proper resources, I would approach this differently.

The easiest option would be to use an existing service such as Endole. They already have a large database of UK companies and VAT information, so instead of trying to discover every VAT number ourselves, we could pay for access to a dataset that already contains this information.

This would probably be the easiest solution if the cost is acceptable.

If we wanted to build our own system instead, I think it would need to use several different sources. We could start with the company name, company number and address, search for the company online, check its website when it has one, look for public documents such as PDFs or invoices, and also check third-party company websites. Any VAT number found this way would then be checked with HMRC before being accepted.

A system like this would be much faster than doing everything manually. It could search many companies at the same time and keep track of where each number came from.

I would also use a proper HMRC API for verification instead of relying on the public checker. This would make the verification process much easier to scale.

I still would not expect it to find every VAT number. Some companies do not have a website, some do not publish their VAT number, and some sources will simply have incomplete or incorrect information.

I would also need to keep checking the data over time. A VAT number found today may no longer be valid later, so the information should not be treated as permanent.

In production, I would track things such as how many companies produce a possible VAT number, how many are accepted by HMRC, which sources produce the best results, and how often a valid number turns out to belong to the wrong company.

The cost would depend heavily on whether we buy an existing database or build the system ourselves. Building our own system would require search API usage, servers, crawling infrastructure, possible proxy costs and time spent maintaining it.

Based on what I found, I would probably start with a commercial service if the goal was to solve the problem quickly and reliably. Building our own system would make more sense if we wanted more control over the data.

The main thing I learned from the research is that the difficult part is not checking whether a VAT number is valid. The difficult part is finding the correct VAT number for the correct company in the first place, especially when doing it for tens of thousands of companies.